

THE 'PRE-LOADED' SHORT INTERMITTENT INTERVALS

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Knowledgeiswatt

January 23, 2026



1. THE SCIENTIFIC BACKGROUND

[In one of our most read KIW Research Note](#) we saw that when the goal is to optimize the aerobic adaptations from high intensity training (HIT, above the second threshold/Critical Power of FTP) training, **the key is to accumulate as much time as possible at a high % of the maximum oxygen consumption (VO₂max).**

Regarding this, we saw that **the short high intensity intermittent intervals** (series of 8-13 min with 3-5 min recovery between where you alternate 15-40s 'on' and 15-30s 'off' in a 1:1 or 2:1 ratio) **are better than traditional long HIT intervals** (3-5 minutes work phases with 1:1 or 2:1 work-recovery ratio). It seems like the capacity to work at a higher intensity despite a similar rate of perceived exertion and total time spent at a high intensity power is what make short intervals superior in promoting aerobic adaptations.

Furthermore, during long intervals, the longer recovery periods—required to sustain the subsequent long work bouts—allow oxygen uptake (VO₂) to drop substantially. As a consequence, the initial portion of each new long interval is spent rebuilding VO₂ rather than

working at a high percentage of VO_2max . This results in a lower time spent at a high fraction of VO_2max across the entire session.

In contrast, the short recovery periods typical of short intermittent intervals prevent VO_2 from declining markedly. Oxygen uptake remains elevated throughout the session, leading to more time accumulated at a high percentage of VO_2max .

This is illustrated in the figure below, which shows oxygen consumption during a long HIT session (LI), consisting of alternating 3-min work bouts with 2-min recovery, and a short HIT session (SI), consisting of alternating 30-s work and 20-s recovery. In both protocols, the work phases are performed at ~120% of critical power, corresponding to Zone 5 in a 5- or 7-zone intensity model.

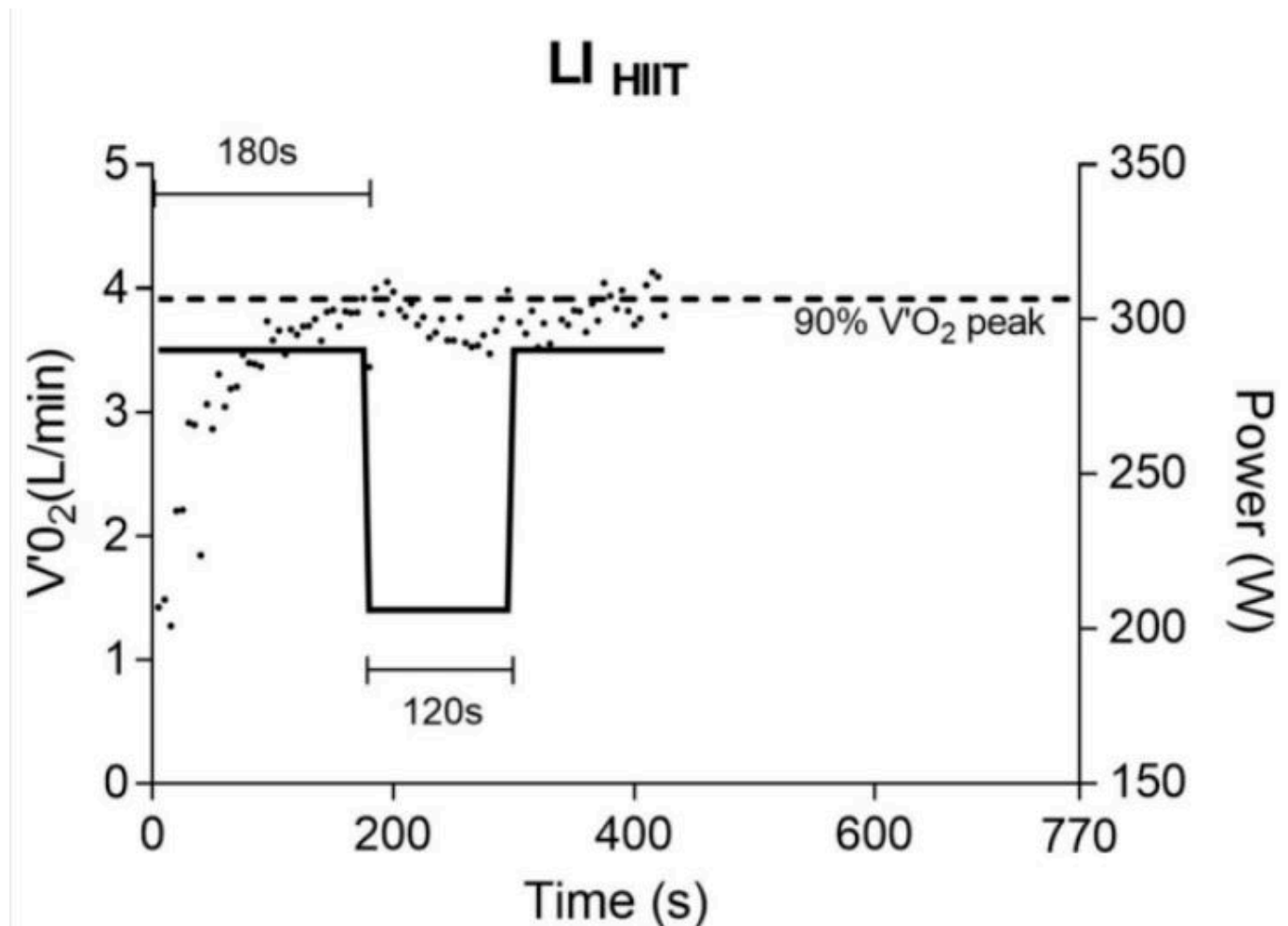


Image adapted from Vaccari et al. (1).

SI HIIT

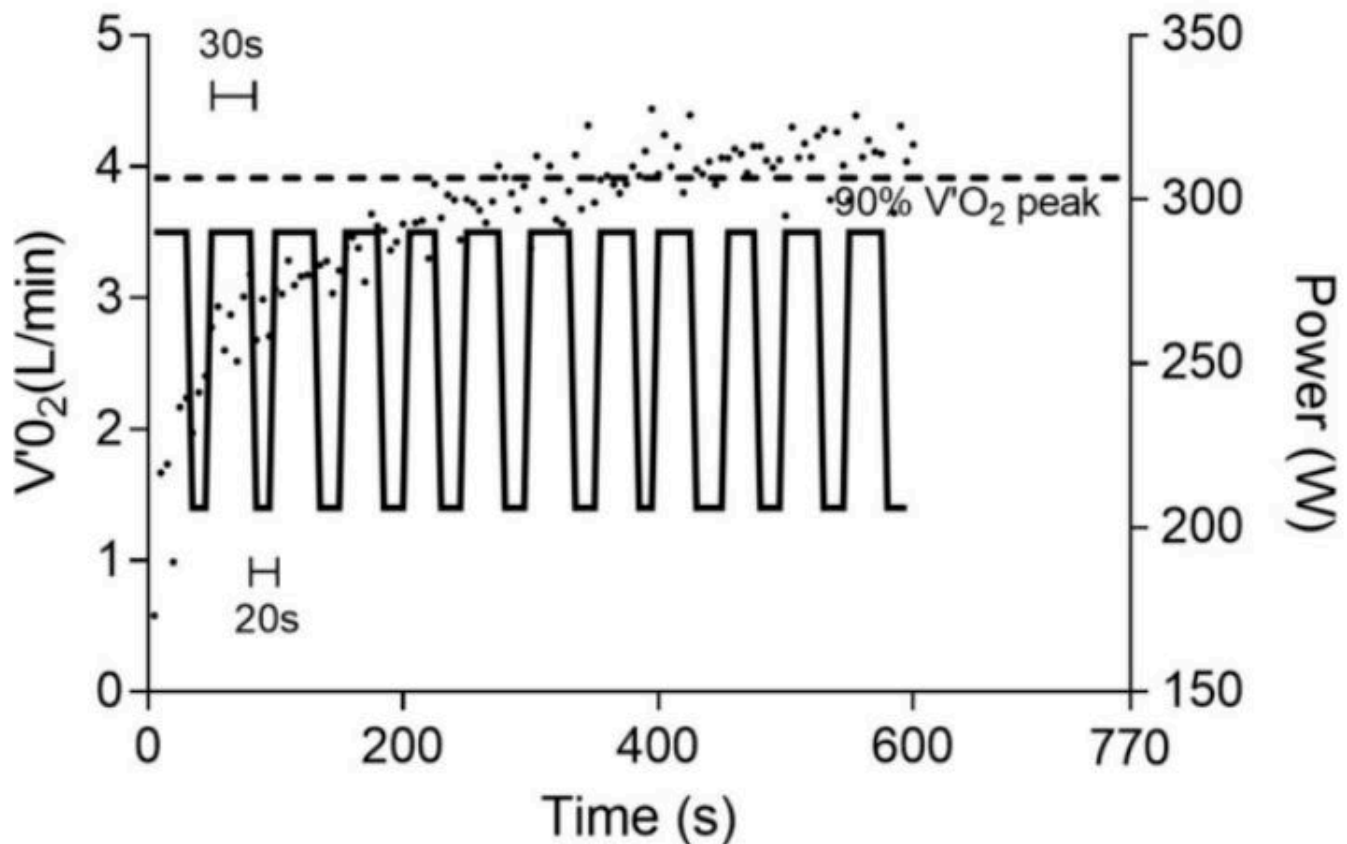


Image adapted from Vaccari et al. (1).

However, it is also evident that in the long-interval format, $\dot{V}O_2$ (oxygen consumption) kinetics—that is, how quickly oxygen uptake rises toward a high fraction of $\dot{V}O_{2\max}$ at the beginning of the interval—is faster than in the short-interval format. For example, after ~180 s, $\dot{V}O_2$ typically reaches ~90% of $\dot{V}O_{2\max}$ during long intervals, whereas during short intermittent intervals it often remains well below this threshold at the same time point.

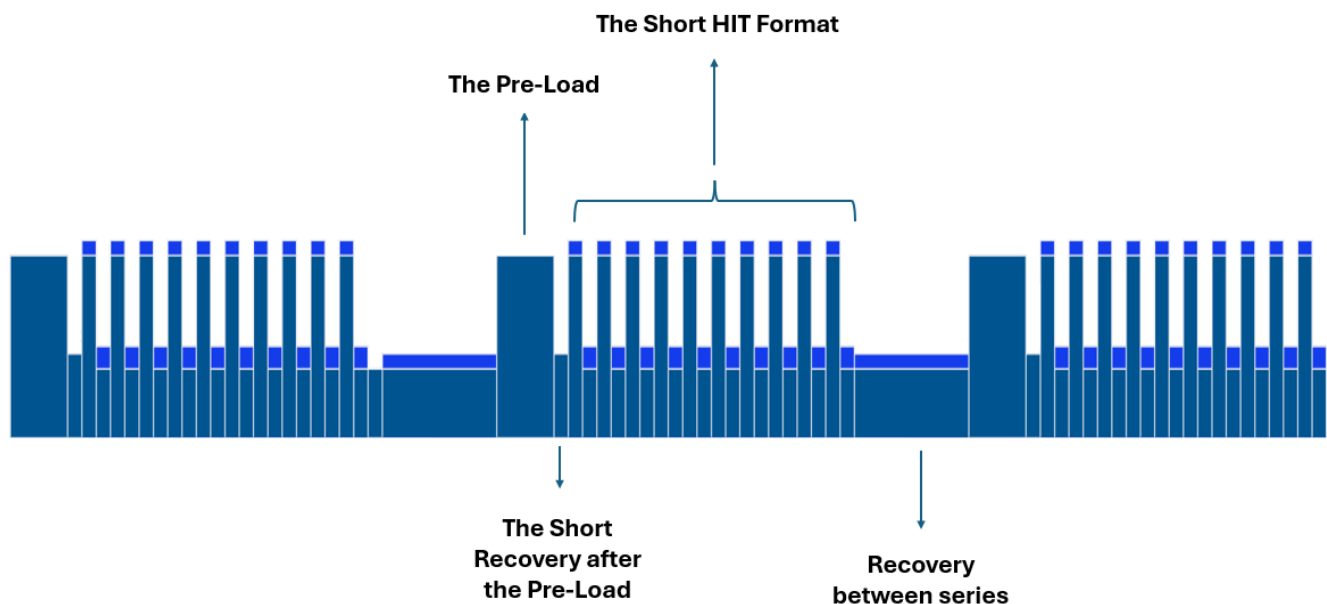
As is often said, training lies somewhere between **art and science**. So, our goal is to combine physiological principles creatively rather than just copy and past studies' protocols. With this in mind, an **optimal strategy can be to exploit both the faster $\dot{V}O_2$ kinetics at the onset of exercise provided by long intervals, while also maximizing the average time spent at a high percentage of $\dot{V}O_{2\max}$ and the accumulated time above 90% of $\dot{V}O_{2\max}$ typical of short intervals.** This can be achieved by starting the workout with one long interval, followed by a short recovery period and then continue with a full short intermittent format.

This combined approach can be called '*pre-loaded short intermittent intervals*'. In line with the physiological principles described above, this format appears to be a highly effective strategy for optimizing aerobic gains induced by high-intensity training.

Here below you can find an example of such a high-intensity workout format!

Please note that, as stated above, here we are talking about optimizing **aerobic** adaptations derived from the HIT sessions. Of course you can still include a 8x1min all-out with 3 min recovery or a 10x15s all-out with 5min recovery HIT sessions in your annual training plan. However the goals of these HIT sessions are to improve anaerobic capacity and neuromuscular power respectively, not aerobic adaptations.

2. THE PRACTICAL EXAMPLE



Please note that the figure above illustrates only the core part of the workout. As with any high-intensity session, a proper warm-up and cool-down should be added before and after the core part, respectively.

Explanation

- **Pre-load interval**

Start with a relatively long pre-load interval: I suggest **2 minutes in Zone 5**, corresponding to ~120% of power at the second threshold (FTP / CP) as a good option. The goal of this first bout is to accelerate VO_2 kinetics reaching a high fraction of $\text{VO}_{2\text{max}}$ earlier than just using the short intermittent intervals in isolation.

- **Short recovery**

Follow the pre-load with a **short recovery**, just long enough to transition into the short intermittent format without experiencing early fatigue. This recovery can range from 30 s (my suggestion) to 1 min, depending on the duration of the pre-load interval (2–3 minutes).

- **Short intermittent intervals**

Short high intensity intermittent intervals are a HIT format in which the work duration is 15sec-1 min and the active-recovery ratio is 1:1 or higher (for example: 3:2 or 2:1). For more details on choosing the interval duration, the intensity selection and how to apply progressive overload using this format, [read this KIWApplied](#).

- **Recovery between series**

Recover 3 to 5 minutes between series.

- **Number of series**

I suggest performing a minimum of 2 and a maximum of 4 series.

The number of series can be progressively increased across the mesocycle to ensure a progressive overload.

- **Applying Progressive overload on this format:** when progressing through the weeks, I would keep constant the pre-load intensity and duration while applying a progression ([using the principles outlined here](#)) only on the short intermittent intervals part.

I have used this workout format in the past and observed that the pre-load interval **doesn't compromise the sustainability** of the short intermittent work that follows. Only a small limitation could emerge when performing **three or four series**: in this case, it may be advisable **not to exceed 10/11 minutes of short intermittent work per series**. Even in this case, when the pre-load is included, the total duration of each series reach **12–13 minutes**, which is already a substantial high-intensity stimulus!

Thanks for reading! Share with your Team!

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REFERENCE:

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1. Vaccari F, Giovanelli N, Lazzer S. High-intensity decreasing interval training (HIDIT) increases time above 90% $\dot{V}O_2$ peak. Eur J Appl Physiol. 2020 Nov;120(11):2397-2405. doi: 10.1007/s00421-020-04463-w. Epub 2020 Aug 11. PMID: 32780251; PMCID: PMC7560936.



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